

Using Marker Based Augmented Reality to teach autistic eating skills

Rahma Bouaziz
Computer Science Department
Taibah University,
Al Madinah, Saudi Arabia,
ReDCAD, University of Sfax, Tunisia
rkammoun@taibahu.edu.sa

Maimounah Alhejaili
Computer Science Department
Taibah University,
Al Madinah, Saudi Arabia,
h.maimounah@gmail.com

Raneem Al-Saedi
Computer Science Department
Taibah University,
Al Madinah, Saudi Arabia,
alsaeidraneem@gmail.com

Abrar Mihdhar
Computer Science Department
Taibah University,
Al Madinah, Saudi Arabia,
Ab.rar.mehdar@gmail.com

Jawaher Alsarrani
Computer Science Department
Taibah University,
Al Madinah, Saudi Arabia,
jawaherahmed.9@gmail.com

Abstract—Autistic children suffer from distraction and difficulty in learning. The research is still ongoing continues to find suitable ways to help autistic children learn and live normally. Recently, the use of digital technologies in supporting children with Autism has increased dramatically. We focus on problems that autistic children face in the learning process. We propose a new learning system based on Augmented Reality overlies digital objects on top of physical cards and rendering them as a 3D object on mobile devices to help in teaching eating food skills using related phrases and sounds. We aim to improve the learning abilities to repeat the correct behavior.

Keywords: Autism Spectrum Disorders (ASD), Augmented Reality, marker-based augmented reality, mobile device, android application.

I. INTRODUCTION

"Autism spectrum disorder (ASD) is a set of neurodevelopmental disorders characterized by a lack of social interaction, verbal and nonverbal communication" [1]. Recently, the use of digital technologies in supporting children with ASD has increased dramatically. Different types of equipment are used: for example, mobile technologies or robots.

They aim to support daily life activities, promote language and social skills learning. The fact explains The use of these technologies corresponds particularly well to the specificities of children with ASD. This developmental disorder manifests itself as difficulties in verbal and nonverbal communication, in terms of expression and comprehension. Difficulties in learning new skills are also observed.

The motivation of developing an application for autistic children is to remedy the lack of Arabic application for autistic children to invest new technologies like Augmented Reality to help those children integrated into Arabic society to improve the learning experience and enabling an attraction of autistic children to the learning process. Since autism children are visual thinkers, the proposed solution aims to enhance their perception and interaction with the real world by projecting virtual objects

in the real world. Also by using the proposed approach, we can reduce the time of learning a new skill.

II. RELATED WORKS

Many researches in helping children with Autism spectrum disorder has been presented. Pradibta et al. [2] proposed an android based application developed by unity 3D for autistic children by using augmented reality flashcard and video audio animation to present a daily prayer for Muslim children. The researcher used Indonesian, Arabic language, and Latin letters in the writings of daily prayers. Pictures and video audio used are downloaded from the Internet. They modify the picture by using Adobe Illustrator to fit the requirements.

Lakshmiprabha, et al. [3] proposed "An Augmented and Virtual Reality System for Training Autistic Children". This research work helps in exploring the use of AR and VR to train children based on applied behavior. The system starts by displaying images and then guides the child to choose one among them and it monitors and strengthens the correct procedure to motivates and encourages the child.

Escobedo, et al. [4] discuss how augmented reality helps keep focus in children with autism: As we know that the autism is related to weakness in attention, memory information processing, and they spend a long time doing a simple task. The authors of this research study how augmented reality can help autistic children by increasing their selective and sustained attention by using the Mobile Object Identification System (Mobis) which rely on "breaks down tasks into smaller components called trials" and to help children keep the focus on a task and have a more positive successful experience. Here, we can see how augmented reality helps redirects the attention of autistic children closing the gap between the physical and digital worlds.

Taryadi et al. [5] [6] present PECS a Picture Exchange Communication System. This system helps in learning children with autism about new pictures with keywords or phrases that are related to these pictures. Also, the system can monitor the

children's actions by tracking their hands by using a projector that can convert all the planar surfaces to a display device and a camera.

Molnár, et al. [7] investigate the use of Augmented Reality in Learning. In their study, the authors discuss the impact of AR programs on the learning process that takes place through the student's own devices. To make learning more effective using several AR platforms like Quiver or HP Reveal (previously Aurasma). Besides, teachers prepare remarkable visual tools or the students can consume this content easily and with more motivation and finally present a survey conducted among students of higher education in Hungary probed the extent, to which augmented reality-based applications such as Pokémon Go.

All of these works have promoted the use of new technologies to learn skills in more effective's ways and different aspects. Since augmented reality based on mobile devices eliminates the need for expensive equipment or the acquisition of new knowledge, the mobile-based technology is designed to be one of the tools of the therapy for children with autism [7]. This research focusing on developing an Augmented Reality mobile application that will help autistic children as well their teachers on learning daily life skills, while making use of the latest technological advances, especially the smartphone area.

III. BACKGROUND

A. Autism:

"Autism spectrum disorder (ASD) is a set of neurodevelopmental disorders characterized by a lack of social interaction, verbal and nonverbal communication in the first 3 years of life. The distinctive social behaviors include avoidance of eye contact, problems with emotional control or understanding the emotions of others, and a markedly restricted range of activities and interests." [1].

B. Augmented Reality

Caudell et al. [8] introduce the term "Augmented Reality" (AR) in 1992. The author uses a prototype as a head-up display headset to know the position of the head in the real world and display virtual content through an Optical see-through real-time system. Azuma et al. [9] propose a more structured definition of augmented reality that does not depend on any used technology. Any augmented reality system must respect these three rules: 3D view, it should combine real and virtual and Interactive in a real-time.

Augmented Reality (AR) is an innovational technology that expands the user's physical world by adding virtual elements in the real world. Augmented reality-based applications work based on two main categories of AR, named as Marker-based AR and Markerless AR.

1) Marker-based AR:

The marker can be a QR code, 2D bar code, or bright color and meaningful pictures. Marker-based AR can be used when a device captures an image by the camera of that device, the software detects it, deduces the location and orientation of the camera, and then overlaps the virtual objects on the screen.

2) Markerless AR

Markerless AR mostly uses location technologies, such as sensor technology that controls the proportional position relationship between virtual elements and the physical world, and Global Positioning System (GPS) which is known as location-based.

When a device camera scans a physical world, this inbuilt GPS software detects the location of the device. A sensor like an accelerometer and gyroscope of the device record this location and orientation of the device. This allows the application to offer data, related to that specific location. Then, rendering the virtual elements on the physical world at a specific location [10] [11].

In this research, we preferred to use the marker-based AR approach because the most common way of learning used with autistic children is by using cards, so we preferred to develop the regular method using marker-based AR technology.

IV. PROPOSED APPLICATION

The concept proposed in this Arabic application is the use of interactive cards for ASD that can be rendering the 3D animation when detect the image features in the cards by using the camera of mobile devices. Besides, this approach enables children to see the skill depicted in the card in the form of a 3D character video shown on the mobile device screen. It is expected that the interactive card can attract children with ASD attention better than using the traditional learning method. We aim that the proposed application helps all the teachers save their time and provide a way to helping autistic children in a fun way using augmented reality. Furthermore, the marker and the animations are developed generally clear fully and easy to be understood for the autistic children.

Cards used in the proposed application are designed with specific software to suit the recommendation of the therapist that considers may motivate the ASD children. We can associate sounds, a word, and all the related requirements. The character's picture is shown in Fig. 1.



Fig. 1. Card for eating skill

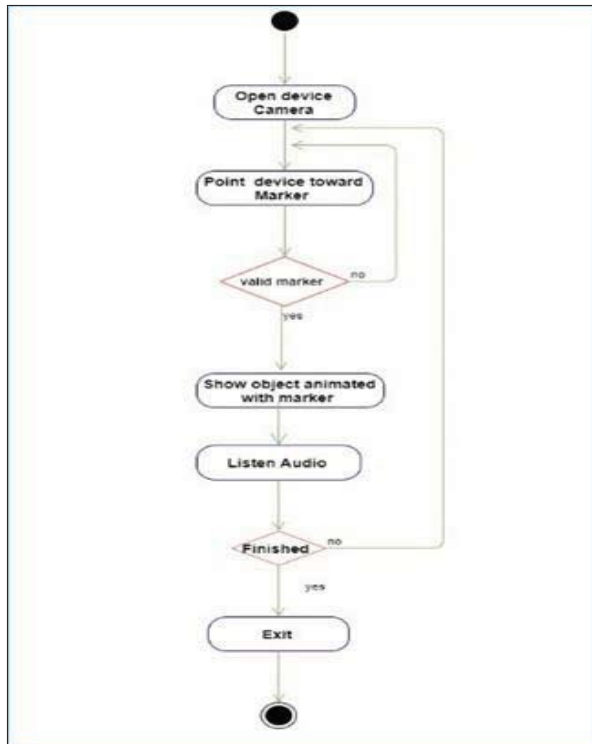


Fig. 2. Activity diagram of the proposed system.

The Activity diagram is a dynamic diagram that shows the sequences of steps to be executed in the proposed application. The interactive card application starts from the mobile device or tablet. It opens and shows anything the camera captures in real-time. Every time the camera finds the marker previously loaded in the session, the application recognizes it and displays on the screen the animated object and sound related to the marker as the current camera capture, as shown in Fig. 2.

Vuforia Engine [12] as an AR platform developed by Parametric Technology Corporation (PTC) company. It is a comprehensive and scalable enterprise platform that provides the appropriate AR technology to customers based on their needs. We used the Vuforia Engine 8.3.8 version (Ptc, 2019). to upload the markers (images). The Vuforia detects the features of the markers and saves these features. Then when the user scans the cards the camera will detect the fracture and check for them in the Vuforia, then shown the 3D animation video. The Fig. 3 shows the steps of application.

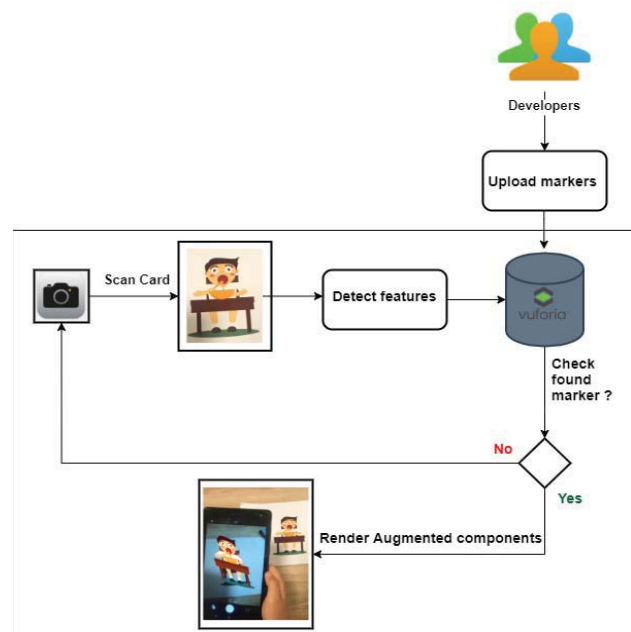


Fig.3. The steps of the application

When the scanned card appears on the screen, the sound loaded and the word will be reproduced as well. It is expected to attract the autistic child with the superposed AR image as shown in Fig. 4.

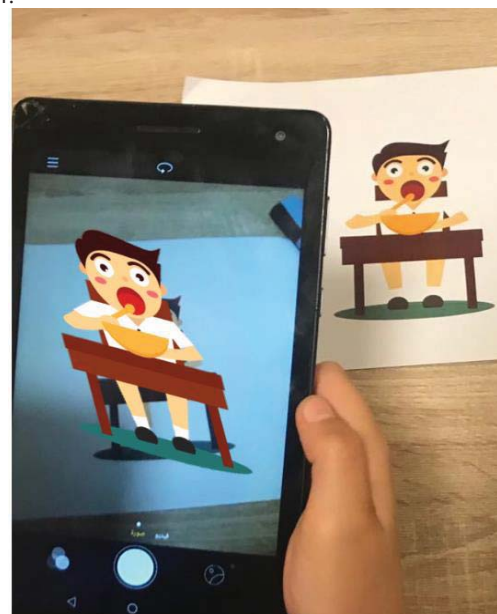


Fig.4. The superposition of the AR object

The gallery of images related to situations to which the card refers is accessible via the photo camera icon in the proposed application (see Fig. 5).



Fig. 5. Gallery of images.

V. DEMO

The following QR code stands for showing a demo for one or more videos of related situations. The application was tested in an android device example Galaxy S10.



Fig. 6. QR code of the Demo of Galaxy S10.

VI. CONCLUSION AND FUTURE WORK

The proposed research proposes an approach that is interested in helping autistic children in Arabic society learn daily life skills using a mobile device. We aim to achieve an

efficient solution that keeps up with the latest technological trends and standards by using AR.

As future work, we aim to expand the gallery of cards of daily life skills and use multiple Arabic dialects. Besides, we promote to enable the users to print the card from the gallery of cards and allow parents to receive an e-mail notification about the progress of their children.

REFERENCES

- [1] Park, H. R., Lee, J. M., Moon, H. E., Lee, D. S., Kim, B. N., Kim, J., Kim, D. G., & Paek, S. H. (2016). A short review on the current understanding of autism spectrum disorders. *Experimental Neurobiology*, 25(1), 1–13. <https://doi.org/10.5607/en.2016.25.1.1>
- [2] Pradibta, H., & Wijaya, I. D. (2017). Designing AR daily prayers for children with ASD. *Proceedings - 2017 International Seminar on Application for Technology of Information and Communication: Empowering Technology for a Better Human Life, ISEmantic 2017*, 2018-Janua, 100–103. <https://doi.org/10.1109/ISEMANTIC.2017.8251851>
- [3] Lakshmiprabha, N. S., Santos, A., Mladenov, D., & Beltramello, O. (2014). An augmented and virtual reality system for training autistic children. *ISMAR 2014 - IEEE International Symposium on Mixed and Augmented Reality - Science and Technology 2014, Proceedings*, 277–278. <https://doi.org/10.1109/ISMAR.2014.6948448>
- [4] Edwards-Stewart, A., Hoyt, T., & Reger, G. M. (2016). Classifying different types of augmented reality technology Edward-Stewart, A., Hoyt, T., & Reger, G. M. (2016). Classifying different types of augmented reality technology. *Annual Review of CyberTherapy and Telemedicine*, 14(January), 199–202. *Annual Review of CyberTherapy and Telemedicine*, 14(January), 199–202.
- [5] Taryadi, & Kurniawan, I. (2016). Multimedia Augmented Reality With Picture Exchange Communication System for Autism Spectrum Disorder. *International Journal of Computer Science and Technology*, 7(4), 102–106.
- [6] Taryadi, & Kurniawan, I. (2018). The improvement of autism spectrum disorders on children communication ability with PECS method Multimedia Augmented Reality-Based. *Journal of Physics: Conference Series*, 947(1). <https://doi.org/10.1088/1742-6596/947/1/012009>
- [7] Molnár, G., Szűts, Z., & Biró, K. (2018). Use of augmented reality in learning. *Acta Polytechnica Hungarica*, 15(5), 209–222. <https://doi.org/10.12700/APH.15.5.2018.5.12>
- [8] Caudell, T. P., & Mizell, D. W. (1992). Augmented reality: An application of heads-up display technology to manual manufacturing processes. 659–669.
- [9] Azuma, R. T., & Others. (1997). A survey of augmented reality. *Presence*, 6, 355–385.
- [10] Liang, S. (2016). Design Principles of Augmented Reality Focusing on the Ageing Population. *BCS HCI*.
- [11] C.P.Cheng, J., Chen, K., & Chen, W. (2017). Comparison of marker-based AR and markerless AR : a case study on indoor decoration system. *Lean and Computing in Construction Congress*, 2.
- [12] Ptc. (2019). Vuforia. <https://www.ptc.com/en/products/augmented-reality/vuforia>